

FINAL EXAM—SOLUTIONS

ECON301—ECONOMETRICS I

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Instructions. Read these instructions carefully! The exam has a total of 100 points. All questions have the same weight (25 points). Read the problems carefully. Write your answers neatly. Please number each page you use. Scan or take a picture of each page of your exam and **combine them into a pdf**. You can use the following free utility for this purpose: <https://combinepdf.com/>. **Your pdf should be legible.** Please submit your file **through the link in Moodle**. **I will not accept submissions that are not through Moodle.** You have until April 30 at 6 PM to submit your exam.

1. The following table reports the results of three regressions of hourly earnings (*AHE*) on a number of regressors using a survey of full time workers in U.S. in 1998. The highest educational achievement for each worker was either a high school diploma or a bachelor's degree. Workers' ages ranged from 25 to 34 years. The data set also contained information on the region of the country where the person lived.

Regressor	(1)	(2)	(3)
College (X_1)	5.46 (0.21)	5.48 (0.21)	5.44 (0.21)
Female (X_2)	-2.64 (0.20)	-2.62 (0.20)	-2.62 (0.20)
Age (X_3)		0.29 (0.04)	0.29 (0.04)
Northeast (X_4)			0.69 (0.3)
Midwest (X_5)			0.60 (0.28)
South			-0.27 (0.26)
Intercept	12.69 (0.14)	4.40 (1.05)	3.75 (1.06)
Summary Statistics			
Residual standard error	6.27	6.22	6.21
R^2	0.176	0.190	0.194
$\bar{R}^2$			
n	4000	4000	4000

- *AHE* = average hourly earnings (in 1998 dollars)
- *College* = binary variable (1 if college, 0 if high school)
- *Female* = binary variable (1 if female, 0 if male)
- *Age* = age (in years)
- *Northeast* = binary variable (1 if Region = Northeast, 0 otherwise)
- *Midwest* = binary variable (1 if Region = Midwest, 0 otherwise)
- *South* = binary variable (1 Region = South, 0 otherwise)
- *West* = binary variable (1 Region = West, 0 otherwise)

- Compute $\bar{R}^2$ for every regression.
- Consider regression (1). Do workers with college degree earn more, on average, than workers with only high school degree? How much more?
- Consider regression (1). Do men earn more than women on average? How much more?
- Consider regression (2). Is Age an important determinant of earnings? Explain.
- Consider regression (2). John is a 31-year-old male without a college degree while Bob is a 24 year-old male with college degree. Predict John's and Bob's earnings.
- Consider regression (3). Do there appear to be important regional differences? Why or why not?
- Consider regression (3). Why was *West* excluded? What would happen if it was included?

Solution

- (a) $\bar{R}_{(1)}^2 = 1 - \frac{4000-1}{4000-2-1}(1 - R_{(1)}^2) = 1 - \frac{3999}{3997}(1 - 0.176) = 0.1756$, $\bar{R}_{(2)}^2 = 1 - \frac{4000-1}{4000-3-1}(1 - R_{(2)}^2) = 1 - \frac{3999}{3996}(1 - 0.190) = 0.1894$ and $\bar{R}_{(3)}^2 = 1 - \frac{4000-1}{4000-6-1}(1 - R_{(3)}^2) = 1 - \frac{3999}{3993}(1 - 0.194) = 0.1928$.
- (b) On average, a worker with a college degree earns 5.46/hour more for than workers without the degree.
- (c) On average, male workers earn 2.64/hour more than female workers.
- (d) Yes, age is an important determinant of earnings. The t -statistic is $0.29/0.04 = 7.25$, implying that the coefficient on age is statistically significant at the 1% level. In terms of the economic impact, note that 0.29/hour is equivalent to $0.29 \times 40 \times 4 \times 12 = \$556.8/\text{year}$. For each additional year the worker ages she gets 556.8 additional dollars on average.
- (e) John's expected earnings are $4.40 + 5.48 \times 0 - 2.62 \times 0 + 0.29 \times 31 = \13.39 per hour. Bob's expected earnings are $4.40 + 5.48 \times 1 - 2.62 \times 0 + 0.29 \times 24 = \16.84 per hour.
- (f) First, note that, except for South, the coefficients on the regional variables are statistically significant at $\alpha = 0.05$ significance level. On average, workers in the Northeast earn \$0.69 more per hour than workers in the West, workers in the Northeast earn \$0.60 more per hour than workers in the West, and workers in the South earn \$0.27 less than workers in the West, all controlling for other variables in the regression. Then, we can claim that there are important regional differences.
- (g) The regressor *West* is omitted to avoid perfect multicollinearity. If *West* is included, then the intercept can be written as a perfect linear function of the four regional regressors and the OLS method will fail. This is the so called *dummy variable trap* problem.

2. Assume that the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ satisfies the Gauss-Markov assumptions and let $\hat{\boldsymbol{\beta}}$ be the OLS estimator of $\boldsymbol{\beta}$. Let $\mathbf{Z} = \mathbf{G}(\mathbf{X})$ be a $n \times (k+1)$ matrix function of $\mathbf{X}$ and assume that $\mathbf{Z}'\mathbf{X}$ (a $(k+1) \times (k+1)$ matrix) is nonsingular. Define a new estimator of $\boldsymbol{\beta}$ by $\tilde{\boldsymbol{\beta}} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}$.

- (a) Show that $E(\tilde{\boldsymbol{\beta}}|\mathbf{X}) = \boldsymbol{\beta}$, so that $\tilde{\boldsymbol{\beta}}$ is also unbiased conditional on $\mathbf{X}$.
- (b) Find $V(\tilde{\boldsymbol{\beta}}|\mathbf{X})$. Make sure to show that your answer is a symmetric $(k+1) \times (k+1)$, matrix.
- (c) Which estimator do you prefer, $\hat{\boldsymbol{\beta}}$ or $\tilde{\boldsymbol{\beta}}$?

Solution

- (a) $E(\tilde{\boldsymbol{\beta}}|\mathbf{X}) = E[(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}|\mathbf{X}] = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$ where the next to last inequality comes from the fact that $\mathbf{Z}$ is a function of $\mathbf{X}$, and therefore, when conditioning on $\mathbf{X}$, it goes outside the conditional expectation. Using the LIE we conclude that $E(\tilde{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and therefore, $\tilde{\boldsymbol{\beta}}$ is unbiased.
- (b) We have:

$$\begin{aligned} V(\tilde{\boldsymbol{\beta}}|\mathbf{X}) &= V[(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}|\mathbf{X}] = [(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'] V(\mathbf{y}|\mathbf{X}) [(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}]' \\ &= [(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'] \sigma^2 \mathbf{I} [(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}]' = \sigma^2 (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}' \mathbf{Z} (\mathbf{X}'\mathbf{X})^{-1}. \end{aligned}$$

Notice that indeed, $V(\tilde{\boldsymbol{\beta}}|\mathbf{X})$ is a symmetric matrix, since it is equal to its transpose.

- (c) The estimator $\tilde{\boldsymbol{\beta}}$ is linear in $\mathbf{y}$ and, as is shown in the first item, it is unbiased. Since the Gauss-Markov assumptions hold, the OLS estimator, $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator. Thus, we are guaranteed that its variance-covariance matrix is "smaller" in the matrix sense than $V(\tilde{\boldsymbol{\beta}}|\mathbf{X})$.^a Therefore, we prefer the OLS estimator.

^a $V(\tilde{\boldsymbol{\beta}}|\mathbf{X}) - V(\hat{\boldsymbol{\beta}}|\mathbf{X})$ is a positive semidefinite matrix.

3. Consider the following population model

$$y = \mu + \varepsilon$$

where μ is a constant, i.e., the population mean, and ε is a random variable such that $\varepsilon \sim N(0, \sigma^2)$. Suppose that you have a random sample of size n of observations on y , that is, $(y_i)_{i=1, \dots, n}$.

- (a) Let $\bar{Y}$ be the sample mean. Show that $\bar{Y}$ is the *best linear unbiased estimator* of μ . *Hint: write the model in vector form and notice that it is just an extremely simple version of the multiple linear model.*
- (b) Define the random variable $W = \frac{\bar{Y} - \mu}{\sqrt{\sigma^2/n}}$. Show that $W \sim N(0, 1)$. *Hint: Use the previous hint together with the results on the multinormal distribution.*
- (c) Define the sample variance as $S^2 = \frac{\sum_{i=1}^n (y_i - \bar{Y})^2}{n-1}$. Show that S^2 is an unbiased estimator of σ^2 . *Hint: Once again, just interpret this whole exercise as a special case of a multiple linear model.*
- (d) Show that $\frac{\sum_{i=1}^n (y_i - \bar{Y})^2}{\sigma^2} \sim \chi^2(n-1)$, i.e., a *chi squared* distribution with $n-1$ degrees of freedom. *Hint: same tip applies, interpret the model as a multiple linear regression model and use the multinormal distribution results.*
- (e) Define the random variable $T = \frac{\bar{Y} - \mu}{\sqrt{S^2/n}}$. Show that $T \sim t(n-1)$, i.e., a Student's *t* distribution with $n-1$ degrees of freedom. *Hint: same tip applies, interpret the model as a multiple linear regression model and use the multinormal distribution results.*

Solution

- (a) Let $\mathbf{s} = (1, \dots, 1)'$ be the *summer vector* i.e., the column vector that has ones in every entry. Then the model can be expressed in matrix form as:

$$\mathbf{y} = \mu \mathbf{s} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (1)$$

where $\mathbf{y}$ is a $n \times 1$ vector that contains the observations of y_i , $\boldsymbol{\varepsilon}$ is the $n \times 1$ vector of the disturbances ε_i , which is multinormal. This model satisfies the Gauss-Markov assumptions:

MLR1 The model is linear in the parameter μ .

MLR2 By assumption, we possess a random sample of $\mathbf{y}$.

MLR3 The only explanatory variable is $\mathbf{s}$, which, trivially, is linearly independent (since it is only one vector.) Thus, here, the matrix of observation consists of a single vector, so that its rank is 1.

MLR4 Since $\boldsymbol{\varepsilon} \sim N(0, \sigma^2)$, then $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, which trivially implies that conditional on $\mathbf{s}$, the expectation is also zero.

MLR5 Since $\boldsymbol{\varepsilon} \sim N(0, \sigma^2)$, $V(\boldsymbol{\varepsilon}|\mathbf{s}) = V(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$.

The OLS estimator is $(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'\mathbf{y} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{Y}$. Since the Gauss-Markov assumptions are satisfied, $\bar{Y}$ is BLUE for μ . For the record:

$$E(\bar{Y}) = E[(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'\mathbf{y}] = (\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'\mu = \mu.$$

$$V(\bar{Y}) = V[(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'\mathbf{y}] = [(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}']\sigma^2\mathbf{I}[(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}]' = \sigma^2(\mathbf{s}'\mathbf{s})^{-1} = \frac{\sigma^2}{n}$$

You already knew about the moments of $\bar{Y}$ from a long time ago, so in here I am just proving them again using matrix algebra and the results from the MLR model.

- (b) The model (1) is also a CNR model so that $\mathbf{y} \sim N(\mu \mathbf{s}, \sigma^2 \mathbf{I})$. Hence,

$$\bar{Y} = (\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'\mathbf{y} \sim N(\mu, \sigma^2/n),$$

by proposition D3 in the slides. Thus, standard results from the normal distribution yield that

$$\frac{\bar{Y} - \mu}{\sigma^2/\sqrt{n}} \sim N(0, 1),$$

which is what we wanted to show.

- (c) Let $\mathbf{M} = \mathbf{I} - \mathbf{s}(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'$, i.e., the residual making matrix. Let $\hat{\mathbf{u}}$ be the $n \times 1$ vector of residuals, $\hat{\mathbf{u}} = (y_i - \bar{Y})_{i=1, \dots, n}$. Then $\sum_{i=1}^n (y_i - \bar{Y})^2 = \hat{\mathbf{u}}'\hat{\mathbf{u}} = \mathbf{y}'\mathbf{M}\mathbf{y}$. Thus, using proposition 10 in the slides

$$E(\hat{\mathbf{u}}'\hat{\mathbf{u}}) = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{s}(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}') = \sigma^2(n-1),$$

for details of the previous calculation regarding the trace of $\mathbf{I} - \mathbf{s}(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'$, check PS5 and the slides. Therefore, by the previous result, the expected value of $S^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n-1}$ is indeed σ^2 and, therefore, it is unbiased.

- (d) We need to show that $\hat{\mathbf{u}}'\hat{\mathbf{u}}/\sigma^2 \sim \chi^2(n-1)$. We now that $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. Define $\boldsymbol{\zeta} = (1/\sigma)\boldsymbol{\varepsilon}$ so that $\boldsymbol{\zeta} \sim N(\mathbf{0}, \mathbf{I})$. Rewrite the model $\mathbf{y} = \mu\mathbf{s} + \boldsymbol{\varepsilon}$ as $\mathbf{y} = \mu\mathbf{s} + \sigma\boldsymbol{\zeta}$. Then $\hat{\mathbf{u}} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mu\mathbf{s} + \sigma\boldsymbol{\zeta}) = \sigma\mathbf{M}\boldsymbol{\zeta}$, using the fact that $\mathbf{M}\mathbf{s} = \mathbf{0}$. Thus, $\hat{\mathbf{u}}'\hat{\mathbf{u}} = \sigma^2\boldsymbol{\zeta}'\mathbf{M}\boldsymbol{\zeta}$, and $\hat{\mathbf{u}}'\hat{\mathbf{u}}/\sigma^2 = \boldsymbol{\zeta}'\mathbf{M}\boldsymbol{\zeta}$. Now, notice that the non-random matrix $\mathbf{M}$ is idempotent with rank $n-1$. Then by result Q2 in the slides MultivariateNormal-distribution.pdf we have that $\hat{\mathbf{u}}'\hat{\mathbf{u}}/\sigma^2 = \boldsymbol{\zeta}'\mathbf{M}\boldsymbol{\zeta} \sim \chi^2(n-1)$.
- (e) Let $V = \frac{\sum_{i=1}^n (y_i - \bar{Y})^2}{\sigma^2}$, which from the last item, we know $V \sim \chi^2(n-1)$. We also know from the second item that $W = \frac{\bar{Y} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1)$. Note that

$$\frac{W}{\sqrt{V/(n-1)}} = \frac{\frac{\bar{Y} - \mu}{\sqrt{\sigma^2/n}}}{\sqrt{\frac{\sum_{i=1}^n (y_i - \bar{Y})^2}{\sigma^2} / (n-1)}} = \frac{\frac{\bar{Y} - \mu}{\sqrt{1/n}}}{\sqrt{\frac{\sum_{i=1}^n (y_i - \bar{Y})^2}{n-1}}} = \frac{\frac{\bar{Y} - \mu}{\sqrt{1/n}}}{\sqrt{S^2}} = \frac{\bar{Y} - \mu}{\sqrt{S^2/n}} = T$$

Thus, if we can show that W and V are independent, then by standard properties of the multivariate normal (cf. slide 11 in the multivariate normal slides), $T \sim t(n-1)$, i.e., T has the Student's t distribution with $n-1$ degrees of freedom.

Showing independence of W and V . Notice that W and V are functions respectively of $\bar{Y}$ and $\hat{\mathbf{u}}$. Thus, if we can show that $\bar{Y}$ and $\hat{\mathbf{u}}$ are independent, the proof would be complete. Recall that $\bar{Y} = \mathbf{A}\mathbf{y}$, where $\mathbf{A} = (\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'$. Thus, $\bar{Y} = \mathbf{A}(\mu\mathbf{s} + \sigma\boldsymbol{\zeta}) = \mu + \sigma\mathbf{A}\boldsymbol{\zeta}$, using $\mathbf{A}\mathbf{s} = \mathbf{1}$. Let $\mathbf{t}_1 = (1/\sigma)\hat{\mathbf{u}} = \mathbf{M}\boldsymbol{\zeta}$ and $\mathbf{t}_2 = (1/\sigma)(\bar{Y} - \mu) = \mathbf{A}\boldsymbol{\zeta}$. Since $\mathbf{A}\mathbf{M} = \mathbf{0}$, the conditions of proposition Q3 in the slides MultivariateNormal-distribution.pdf are met, so $\mathbf{t}_1$ and $\mathbf{t}_2$ are independent. So $\hat{\mathbf{u}} = \sigma\mathbf{t}_1$ and $\bar{Y} = \mu + \sigma\mathbf{t}_2$ are independent. Finally, since W and V are functions of $\bar{Y}$ and $\hat{\mathbf{u}}$ respectively, they are also independent.

4. You are hired by the office of the *Akimat* to study whether a tax on liquor has decreased average liquor consumption in Nur-Sultan. You are able to obtain, for a sample of individuals selected at random, the difference in liquor consumption (in ounces) for the years before and after the tax. For person i who is sampled randomly from the population, Y_i denotes the change in liquor consumption. Treat these as a random sample from a $\text{Normal}(\mu, \sigma^2)$ distribution.
- (a) The null hypothesis is that there was no change in average liquor consumption. State this formally in terms of μ .
- (b) The alternative is that there was a decline in liquor consumption; state this formally in terms of μ .
- (c) Now, suppose your sample size is $n = 900$, and you obtain estimates $\bar{Y} = -32.8$ and $S^2 = 217528.96$. Calculate the appropriate test statistic for testing H_0 against H_1 . Do you reject H_0 at a size of 5%? At the size of 1%?

Solution

- (a) $H_0 : \mu = 0$.
- (b) $H_a : \mu < 0$.
- (c) The standard error of $\bar{y}$ is $s/\sqrt{n} = 466.4/30 \approx 15.55$. Therefore, the t -statistic for testing H_0 against H_a is $t^\circ = -32.8/15.55 \approx -2.11$. The one-sided p -value is given by $\Pr(t^\circ \leq -2.11)$, where $t^\circ \sim t(899)$. You can find this probability in any tabulation of the Student's t distribution. The p -value is 0.0176. Because the p -value is below 0.05, we reject H_0 against the one-sided alternative at the 5% level. We do not reject at the 1% level because the p -value is greater than 0.01.